BANGALORE INSTITUTE OF TECHNOLOGY

An Autonomous Institution under Visvesvaraya Technological University, Belagavi
K. R. Road, V. V. Puram, Bengaluru - 560004

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MODULE-WISE QUESTION BANK

Academic Year :	**2026-2027**	**Batch :**	**2024-2028**
Semester & Section :	**V**	**Course Title & Course Code :**	**Theory of Computation (BCS503)**
Credits :	**4**	**Course Coordinator :**	**Prof Lakshmi B N**
Total Contact Hours :	**4**	**Total Hours of Pedagogy:**	
Faculty In-charge(s) :	**Prof. Chaithra G V**	**No. of Modules :**	**1**

Guidelines:

- Questions shall be mapped to the appropriate Course Outcome (CO).
- Bloom's cognitive level shall be indicated as L1 to L6, wherever applicable.
- Question category may be indicated as Conceptual / Numerical / Application / Analysis / Design / Case Study / Programming / Other, as applicable.
- The question bank shall provide adequate coverage of all modules and relevant COs.

Bloom's Taxonomy Reference

L1	L2	L3	L4	L5	L6
Remember	Understand	Apply	Analyse	Evaluate	Create

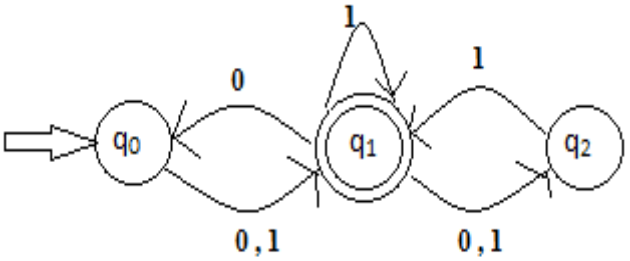
MODULE - 1: Introduction to Automata Theory

Sl. No.	MODULE – 1	CO's	Blooms Levels
1	Define the following with suitable example a. Alphabet b. Power of an alphabet c. Symbols d. Strings e. Languages f. DFA g. NFA	1	1
2	Define Automata Theory? Explain why we need to study automata theory	1	2
3	Explain with an example when we say that a Language is accepted by a Machine?	1	2
4	Explain the applications of Automata Theory.	1	2
5	With a neat diagram explain the Hierarchy of Language Classes.	1	2
6	Outline DFA ,NFA & ε -NFA with an example.	1	2
7	Contrast between DFA,NFA & ε NFA	1	2
8	Build DFA to accept the following language over the alphabet { 0 , 1 } (i) L = { w / w is a even number} (ii) L = { $(0\ 1)^i\ 1^{2j}$ | i>= 1, j >= 1 } (iii) The set of strings either starts or ends or both with 01 (iv) Set of all strings starting with at least two 0's and ending with at least two 1's (v) The strings with consecutive 0's (vi) Set of all strings not containing 110 (vii) Set of all string with exactly three consecutive 0's	2	3

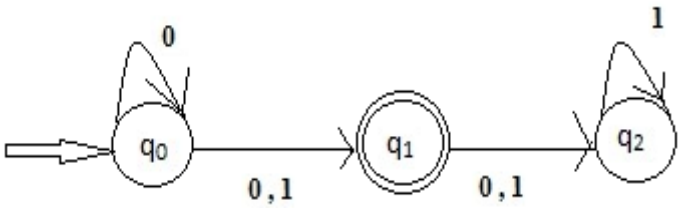
	(viii) Set of all string having substring 001 (ix) Set of all strings such that number of 1's is even and the number of 0's is a multiple of (x) L = { w | w is of even length and begins with 01} (xi) L = { w | w consists of 0's and 1's with 011 as substring } (xii) L = { w | w ∈ 0 (1)^n 2; where n is an odd number}		
9	Build DFA for the following L on ∑ = {a, b} (i) The set of all strings that either begins or ends or both with substring ab (ii) The set of all strings that ends with substring abb (iii) The set of all strings not containing the substring aab (iv) L = { w : |w| mod 5 <> 0} or L = { w : |w| mod 5 = 0} (v) The set of all strings which do not ends with substring abb (vi) L = { a w a | w ∈ { a, b }* } (vii) Set of all strings with at least one 'a' and exactly two b's (viii) L = { w | n_a (w) <= 4; where w ∈ { a, b }* } (ix) L = { w | n_a (w) >= 1 ; n_b (w) = 2 } (x) L = { w | n_a (w) >= 2 ; n_b (w) = 3 } (xi) L = { w | n_a (w) >= 3 ; n_b (w) >= 2 } (xii) L = { w | | w | mod 3 >= | w | mod 2 } (xiii) L = { w | | w | mod 3 = | w | mod 2 where w ∈ ∑* and ∑ = { a } } (xiv) L = { w | w ∈(a + b)* | n_a (w) mod 3 =2 and n_b (w) mod 2 = 1 (xv) Every block of length Five contains at least two a's. Use Extended Transitions of function to trace a string w = aabba (xvi) The number of a's is divisible by 4 and no of b's divisible by 2 (xvii) Set of all strings having odd number of a's and odd number of b's	2	3
10	(i)Build a DFA to accept the binary numbers (i) divisible by 5 (ii) divisible by 3 (ii)Build a DFA to accept the decimal numbers (i) divisible by 5 (ii) divisible by 3	2	3
11	Design ε NFA or NFA for the following languages (i) abc, abd, aacd on ∑ = { a, b, c, d } (ii) { ab, abc}* on ∑ = { a, b, c } (iii)L = { abab^n | n >= 0 } U { aba^n | n >=0 } (iv)L = { aa*(a + b)}	2	3
12	Convert the following NFA to DFA a. | δ | 0 | 1 |	2	3

→q_0	q_0	q_0 , q_1
q_1	q_2	q_2
*q_2	Φ	Φ

b.



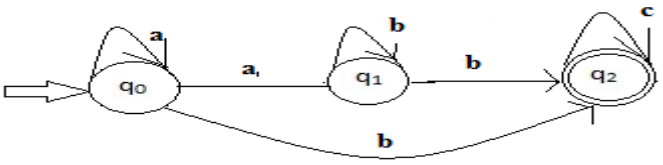
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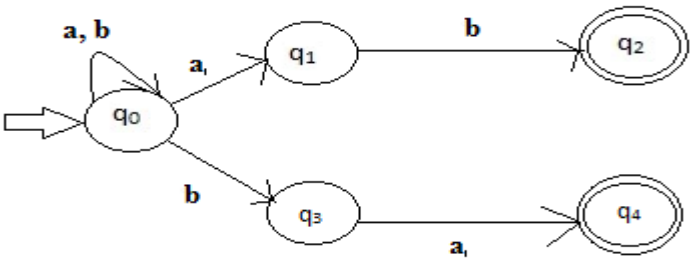
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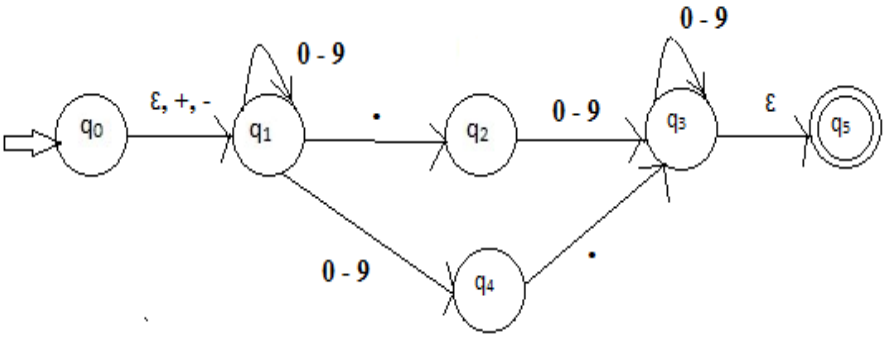
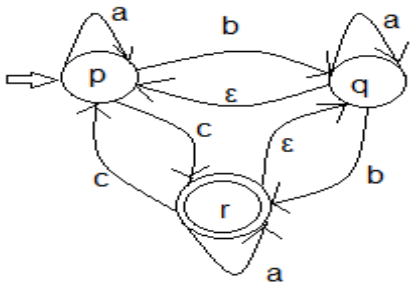
Δ	A	B
q_0	q_0 , q_1	q_0
q_1	-	q_3
q_2	-	-

e.



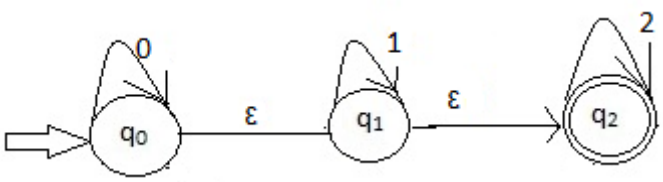
f.



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<tr><td>13</td><td>Convert the following NFA to DFA using “Subset construction scheme”
a.
<table><tr><td>Δ</td><td>0</td><td>1</td></tr><tr><td>→p</td><td>{ p, q }</td><td>{ p }</td></tr><tr><td>q</td><td>{ r }</td><td>{ r }</td></tr><tr><td>r</td><td>{ s }</td><td>Φ</td></tr><tr><td>*s</td><td>{ s }</td><td>{ s }</td></tr></table>b.
 
</td><td>2</td><td>3</td></tr>
<tr><td>14</td><td>What are the Steps involved in converting ε-NFA to DFA Convert the following ε-NFA to DFA
a.
<table><tr><td>Δ</td><td>E</td><td>A</td><td>B</td><td>C</td></tr><tr><td>→p</td><td>Φ</td><td>{ p }</td><td>{ q}</td><td>{ r }</td></tr><tr><td>Q</td><td>{ p }</td><td>{ q}</td><td>{ r }</td><td>Φ</td></tr><tr><td>*r</td><td>{ q}</td><td>{ r }</td><td>Φ</td><td>{ p }</td></tr></table>b.
 

c.</td><td>2</td><td>3</td></tr>
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d.



Δ	E	A	B
→p	{ r }	{ q }	{p, r}
Q	Φ	{ p }	Φ
*r	{p, q}	{ r }	{ p }

e.

Δ	E	A	B	C
→q_0	q_1	q_0, q_1	q_2	-
q_1	q_2	-	q_1, q_2	-
* q_2	-	-	-	q_2

Faculty In-charges

Course Coordinator **Module Coordinator** **IQAC** **Program Coordinator**